

OneDrive > Desktop > Matlab

HW2_CHE321.m HW2Q2_CHE321.m

Command Window

Part (a):

PFR volume for 40% conversion: 71.9 dm^3

CSTR volume for 40% conversion: 24.0 dm^3

Part (b):

PFR and CSTR volumes are approximately equal at $X = 0.20$

Volume at this point $\approx 48.0 \text{ dm}^3$

Part (c):

Maximum conversion in 105 dm^3 CSTR: 0.755 (75.5%)

Part (d):

72 dm^3 PFR $\rightarrow 24 \text{ dm}^3$ CSTR series:

Exit conversion from PFR: 0.221

Final conversion: 0.614 (61.4%)

Part (e):

24 dm^3 CSTR $\rightarrow 72 \text{ dm}^3$ PFR series:

Exit conversion from CSTR: 0.400

Final conversion: 0.900 (90.0%)

Part (f): Plot generated showing conversion and rate vs PFR volume up to 100 dm^3

>> Press **Ctrl** + **Shift** + **P** to generate code with Copilot

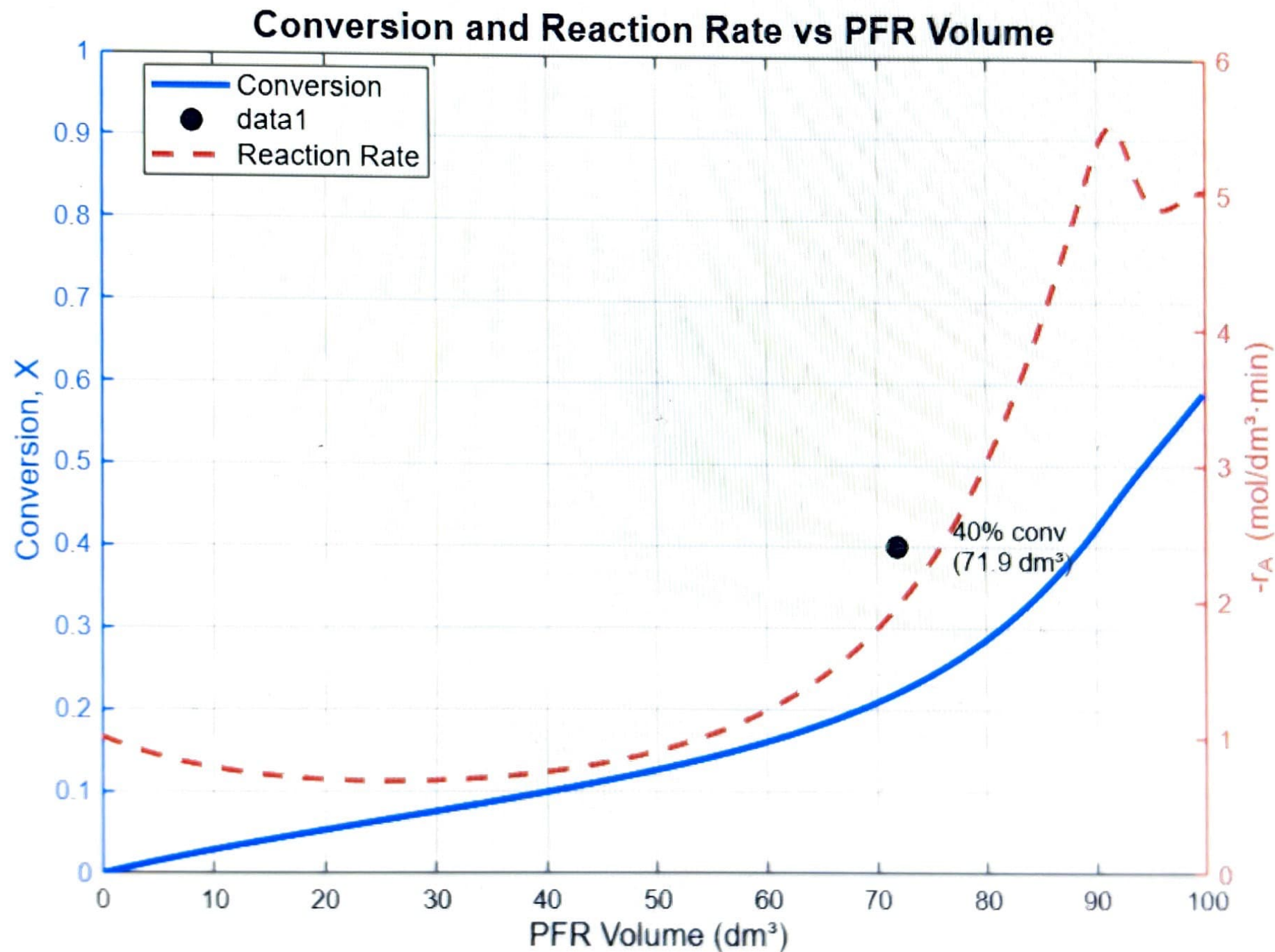
Figure 1

FIGURE

FORMAT

TOOLS

LAYOUT



HW2_CHE321.m x HW2Q2_CHE321.m x +

```

1      clc; clear; close all;
2
3      % Problem 1 - Reactor Design Analysis I
4      % Given experimental data
5      X = [0.0 0.2 0.4 0.45 0.5 0.6 0.8 0.9];
6      rA = [1.00 1.67 5.00 5.50 5.00 5.00 1.25 0.91];
7      FA0 = 300; % mol/min entering molar flow rate
8      Ca0 = 1; % initial concentration (assumed)
9
10     %% Part (a) - Reactor volumes for 40% conversion
11     X_a = 0.4;
12
13     % For PFR: V = FA0 * integral(dX/(-rA)) from 0 to X
14     data_points = find(X <= X_a);
15     V_PFR_40 = trapz(X(data_points), FA0 ./ rA(data_points));
16
17     % For CSTR: V = FA0 * X / (-rA)
18     rA_at_40 = interp1(X, rA, X_a, 'spline');
19     V_CSTR_40 = FA0 * X_a / rA_at_40;
20
21     fprintf('Part (a):\n');
22     fprintf('PFR volume for 40% conversion: %.1f dm³\n', V_PFR_40);
23     fprintf('CSTR volume for 40% conversion: %.1f dm³\n', V_CSTR_40);
24     fprintf('\n');
25
26     %% Part (b) - Finding where PFR and CSTR volumes are equal
27     % Calculate volumes for each conversion
28     V_PFR_array = zeros(size(X));
29     for i = 2:length(X)
30         V_PFR_array(i) = trapz(X(1:i), FA0 ./ rA(1:i));
31     end
32
33     V_CSTR_array = FA0 * X ./ rA;
34
35     % Find minimum difference
36     volume_diff = abs(V_PFR_array - V_CSTR_array);
37     [~, equal_idx] = min(volume_diff(2:end)); % skip X=0
38     equal_idx = equal_idx + 1; % adjust for skipped first element
39
40     fprintf('Part (b):\n');
  
```

HOME

PLOTS

APPS

EDITOR

PUBLISH

VIEW

New

Open

Save

Print

Compare

Go To

Find

Bookmark

Copilot

Refactor

Profiler

Code Issues

Debugger

Section Break

Run and Advance

Run to End

Run

Step

FILE

NAVIGATE

CODE

ANALYZE

SECTION

RUN

OneDrive > Desktop > Matlab

HW2_CHE321.m ×

HW2Q2_CHE321.m ×

```

40 fprintf('Part (b):\n');
41 fprintf('PFR and CSTR volumes are approximately equal at X = %.2f\n', X(equal_idx));
42 fprintf('Volume at this point ≈ %.1f dm³\n', V_PFR_array(equal_idx));
43 fprintf('\n');
44
45 %% Part (c) - Maximum conversion in 105 dm³ CSTR
46 V_target_c = 105;
47
48 % Create fine grid for interpolation
49 X_fine = 0:0.001:0.95;
50 rA_fine = interp1(X, rA, X_fine, 'spline', 'extrap');
51
52 % Calculate required CSTR volumes
53 V_CSTR_fine = FA0 * X_fine ./ rA_fine;
54
55 % Find conversion that gives closest to target volume
56 [~, max_idx] = min(abs(V_CSTR_fine - V_target_c));
57 X_max_CSTR = X_fine(max_idx);
58
59 fprintf('Part (c):\n');
60 fprintf('Maximum conversion in 105 dm³ CSTR: %.3f (%.1f%%)\n', X_max_CSTR, X_max_CSTR*100);
61 fprintf('\n');
62
63 %% Part (d) - Series: 72 dm³ PFR followed by 24 dm³ CSTR
64 V_PFR_d = 72;
65 V_CSTR_d = 24;
66
67 % Step 1: Find exit conversion from PFR
68 X_grid = 0:0.001:0.9;
69 rA_grid = interp1(X, rA, X_grid, 'spline');
70 V_PFR_grid = zeros(size(X_grid));
71 for i = 2:length(X_grid)
72     V_PFR_grid(i) = trapz(X_grid(1:i), FA0 ./ rA_grid(1:i));
73 end
74
75 [~, pfr_idx] = min(abs(V_PFR_grid - V_PFR_d));
76 X_exit_PFR = X_grid(pfr_idx);
77
78 % Step 2: CSTR after PFR
79 % For CSTR: V = FA0*(X_out - X_in)/(rA_out)

```



New



Open



Save



Print



Compare



Go To



Find

▼

Bookmark

▼



Copilot



Refactor



Code Issues

▼

Debugger

▼



Profiler

▼

Code Issues

▼

ANALYZE



Run

Section

▼



Section Break

▼

Run and Advance

▼

Run to End

▼



Run

▼

FILE

NAVIGATE

CODE

SECTION

> OneDrive > Desktop > Matlab



HW2_CHE321.m x HW2Q2_CHE321.m x +

```

80 % Solve: 24 = 300*(X_out - X_exit_PFR)/rA_interp(X_out)
81
82 cstr_func = @(X_out) FA0 * (X_out - X_exit_PFR) / interp1(X, rA, X_out, 'spline') - V_CSTR_d;
83 X_final_d = fzero(cstr_func, X_exit_PFR + 0.1);
84
85 fprintf('Part (d):\n');
86 fprintf('72 dm³ PFR → 24 dm³ CSTR series:\n');
87 fprintf('Exit conversion from PFR: %.3f\n', X_exit_PFR);
88 fprintf('Final conversion: %.3f (0.1f%%)\n', X_final_d, X_final_d*100);
89 fprintf('\n');
90
91 %% Part (e) - Series: 24 dm³ CSTR followed by 72 dm³ PFR
92 % Step 1: Exit conversion from CSTR
93 cstr_alone = @(X_out) FA0 * X_out / interp1(X, rA, X_out, 'spline') - V_CSTR_d;
94 X_exit_CSTR = fzero(cstr_alone, 0.3);
95
96 % Step 2: PFR after CSTR
97 X_range = X_exit_CSTR:0.001:0.9;
98 rA_range = interp1(X, rA, X_range, 'spline');
99 V_PFR_range = cumtrapz(X_range, FA0 ./ rA_range);
100
101 [~, final_idx] = min(abs(V_PFR_range - V_PFR_d));
102 X_final_e = X_range(final_idx);
103
104 fprintf('Part (e):\n');
105 fprintf('24 dm³ CSTR → 72 dm³ PFR series:\n');
106 fprintf('Exit conversion from CSTR: %.3f\n', X_exit_CSTR);
107 fprintf('Final conversion: %.3f (0.1f%%)\n', X_final_e, X_final_e*100);
108 fprintf('\n');
109
110 %% Part (f) - Plot conversion and rate vs PFR volume
111 X_plot = 0:0.005:0.9;
112 rA_plot = interp1(X, rA, X_plot, 'spline');
113
114 V_plot = zeros(size(X_plot));
115 for i = 2:length(X_plot)
116     V_plot(i) = trapz(X_plot(1:i), FA0 ./ rA_plot(1:i));
117 end
118
119 % Only plot up to 100 dm³

```





OneDrive > Desktop > Matlab

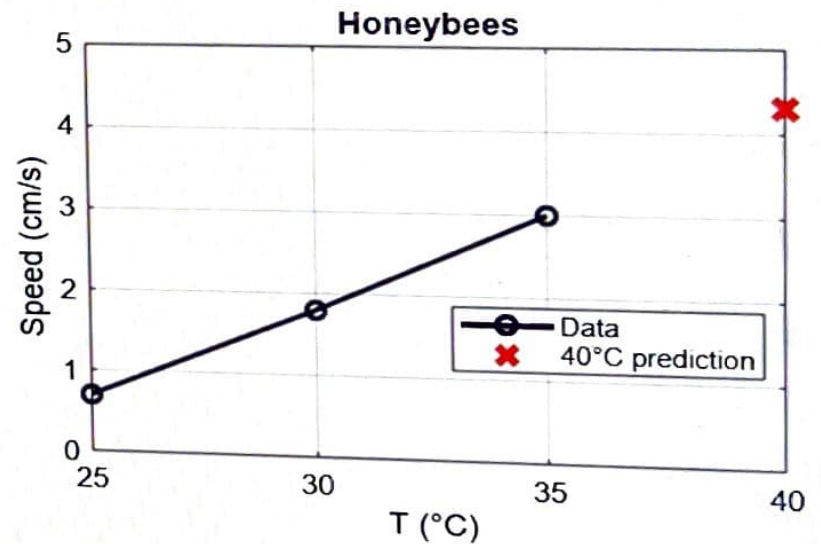
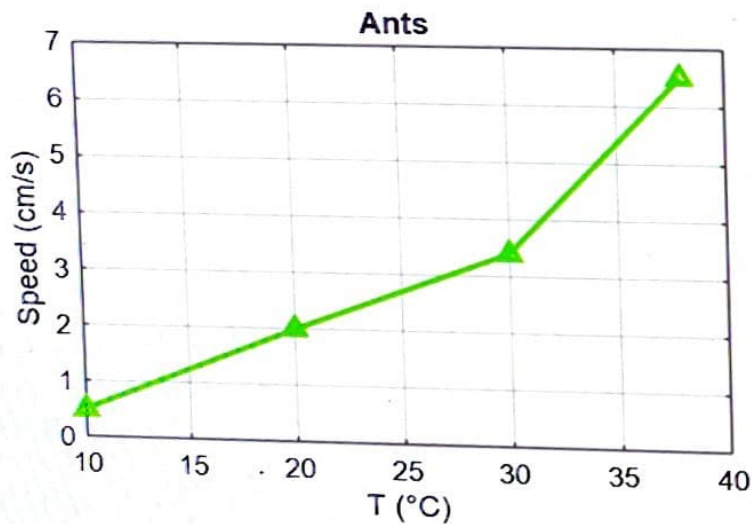
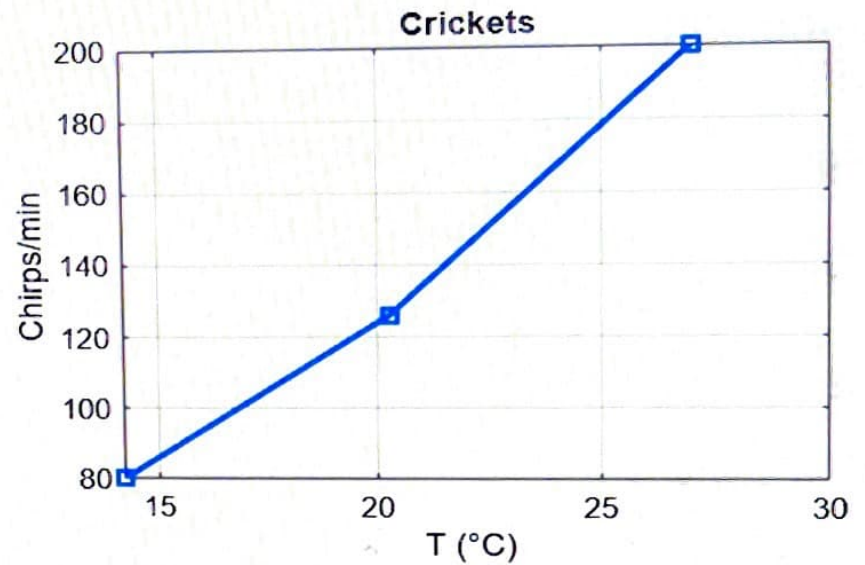
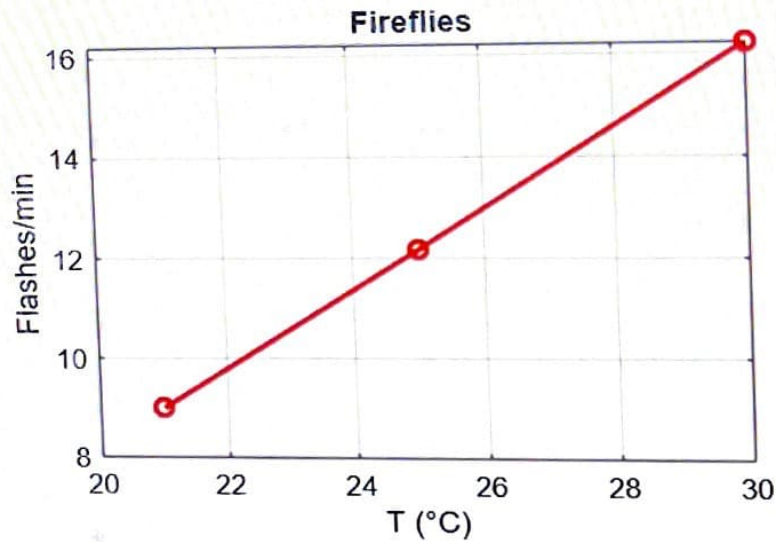
HW2_CHE321.m x HW2Q2_CHE321.m x

```

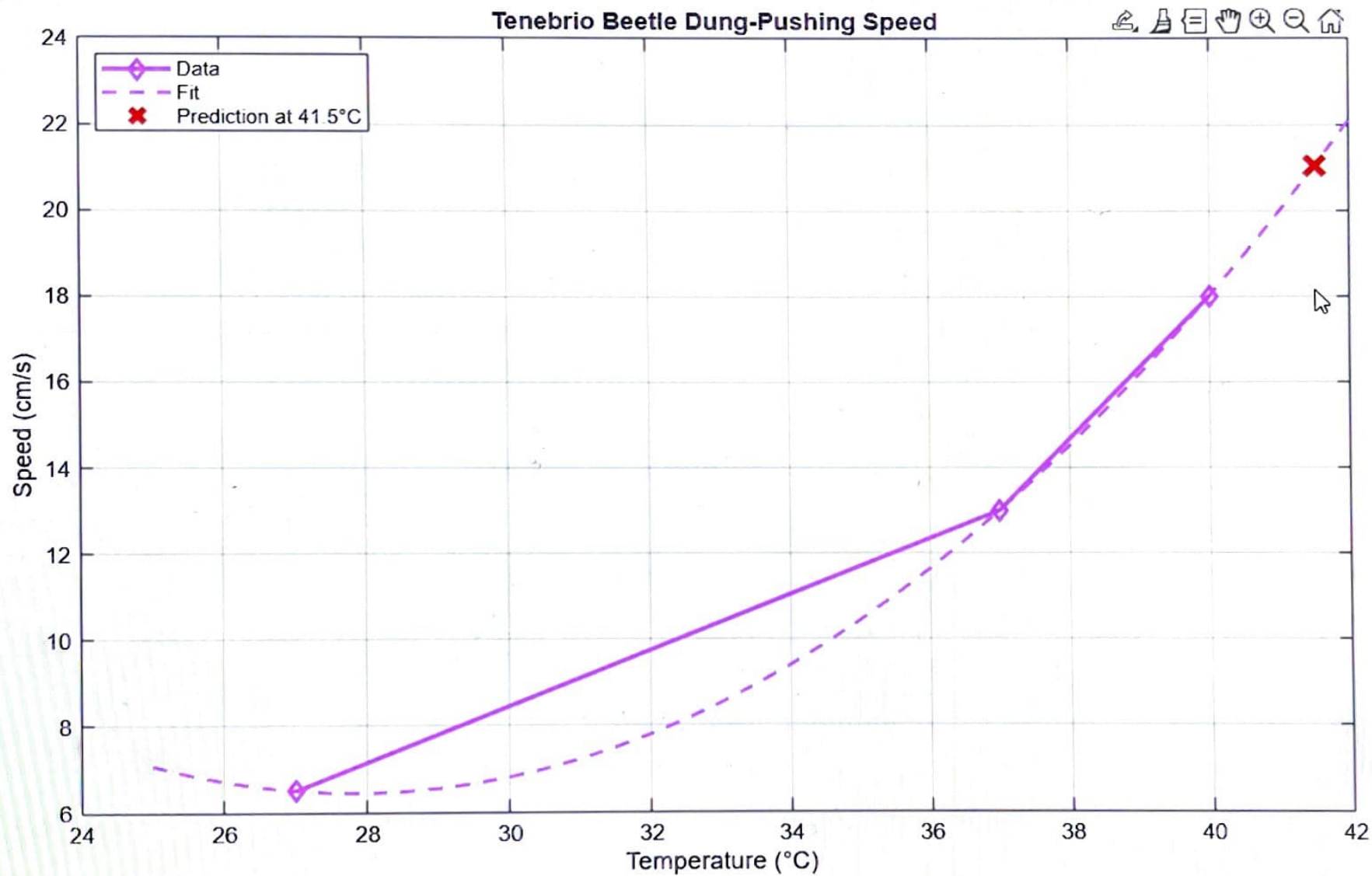
118
119 % Only plot up to 100 dm³
120 valid_range = V_plot <= 100;
121 V_plot = V_plot(valid_range);
122 X_plot = X_plot(valid_range);
123 rA_plot = rA_plot(valid_range);
124
125 figure('Position', [100 100 800 500]);
126 yyaxis left
127 plot(V_plot, X_plot, 'b-', 'LineWidth', 2.5);
128 ylabel('Conversion, X', 'FontSize', 12);
129 ylim([0 1]);
130 set(gca, 'YColor', 'b');
131
132 yyaxis right
133 plot(V_plot, rA_plot, 'r--', 'LineWidth', 2);
134 ylabel('-r_A (mol/dm³·min)', 'FontSize', 12);
135 set(gca, 'YColor', 'r');
136
137 xlabel('PFR Volume (dm³)', 'FontSize', 12);
138 title('Conversion and Reaction Rate vs PFR Volume', 'FontSize', 14);
139 legend({'Conversion', 'Reaction Rate'}, 'Location', 'best', 'FontSize', 11);
140 grid on;
141 xlim([0 100]);
142
143 % print table image
144 hold on;
145 yyaxis left;
146 plot(V_PFR_40, X_a, 'ko', 'MarkerSize', 8, 'MarkerFaceColor', 'k');
147 text(V_PFR_40+5, X_a, sprintf('40% conv\n(%.1f dm³)', V_PFR_40), 'FontSize', 10);
148
149 fprintf('Part (f): Plot generated showing conversion and rate vs PFR volume up to 100 dm³\n');

```

Insect Activity vs Temperature



```
fprintf('Similar exponential-type growth with temperature\n\n');
```



HOME PLOTS APPS EDITOR PUBLISH VIEW

New Open Save Print Compare Go To Find Bookmark Copilot Refactor CODE Profiler Code Issues Debugger ANALYZE Run Section Run and Advance Run to End SECTION Run

FILE > OneDrive > Desktop > Matlab

HW2_CHE321.m x HW2Q2_CHE321.m x +

```
40 fprintf('Part (d) - Experimental improvements:\n');
41 fprintf('Need wider temperature ranges and more data points\n');
42 fprintf('Suggest testing ants: easy to observe, measure running speed\n');
43 fprintf('Temperature range: 5-45°C in 5°C steps\n\n');
44
45 %% Part (e) - Tenebrio beetle analysis
46 beetle_T = [27, 37, 40]; % °C
47 beetle_speed = [6.5, 13, 18]; % cm/s
48
49 beetle_fit = polyfit(beetle_T, beetle_speed, 2);
50 speed_41p5 = polyval(beetle_fit, 41.5);
51
52 fprintf('Part (e) - Beetle analysis:\n');
53 fprintf('Predicted speed at 41.5°C: %.1f cm/s\n', speed_41p5);
54 fprintf('Reference value: 19.2 cm/s\n\n');
55
56 %% Plotting results
57 figure('Position', [100, 100, 900, 600]);
58
59 subplot(2,2,1);
60 plot(firefly_T, firefly_rate, 'ro-', 'LineWidth', 2);
61 title('Fireflies'); xlabel('T (°C)'); ylabel('Flashes/min'); grid on;
62
63 subplot(2,2,2);
64 plot(cricket_T, cricket_rate, 'bs-', 'LineWidth', 2);
65 title('Crickets'); xlabel('T (°C)'); ylabel('Chirps/min'); grid on;
66
67 subplot(2,2,3);
68 plot(ant_T, ant_vel, 'g^-', 'LineWidth', 2);
69 title('Ants'); xlabel('T (°C)'); ylabel('Speed (cm/s)'); grid on;
70
71 subplot(2,2,4);
72 plot(bee_T(valid_bee), bee_vel(valid_bee), 'ko-', 'LineWidth', 2);
73 hold on;
74 plot(40, vel_40C, 'rx', 'MarkerSize', 10, 'LineWidth', 3);
75 title('Honeybees'); xlabel('T (°C)'); ylabel('Speed (cm/s)');
76 legend('Data', '40°C prediction', 'Location', 'best'); grid on;
77
78 sgtitle('Insect Activity vs Temperature');
79
```

HOME

PLOTS

APPS

EDITOR

PUBLISH

VIEW



New



Open



Save



Print



Compare

FILE



Go To



Find



Bookmark

NAVIGATE



Copilot



Refactor

CODE



Profiler



Code Issues



Debugger

ANALYZE



Run and Advance



Section Break



Run to End

SECTION

> OneDrive > Desktop > Matlab

HW2_CHE321.m x HW2Q2_CHE321.m x +

```
67 subplot(2,2,3);
68 plot(ant_T, ant_vel, 'g^-', 'LineWidth', 2);
69 title('Ants'); xlabel('T (°C)'); ylabel('Speed (cm/s)'); grid on;
70
71 subplot(2,2,4);
72 plot(bee_T(valid_bee), bee_vel(valid_bee), 'ko-', 'LineWidth', 2);
73 hold on;
74 plot(40, vel_40C, 'rx', 'MarkerSize', 10, 'LineWidth', 3);
75 title('Honeybees'); xlabel('T (°C)'); ylabel('Speed (cm/s)');
76 legend('Data', '40°C prediction', 'Location', 'best'); grid on;
77
78 sgtitle('Insect Activity vs Temperature');
79
80 % Additional beetle plot
81 figure;
82 plot(beetle_T, beetle_speed, 'md-', 'LineWidth', 2);
83 hold on;
84 T_range = 25:0.5:42;
85 speed_range = polyval(beetle_fit, T_range);
86 plot(T_range, speed_range, 'm--', 'LineWidth', 1.5);
87 plot(41.5, speed_41p5, 'rx', 'MarkerSize', 12, 'LineWidth', 3);
88 title('Tenebrio Beetle Dung-Pushing Speed');
89 xlabel('Temperature (°C)'); ylabel('Speed (cm/s)');
90 legend('Data', 'Fit', 'Prediction at 41.5°C', 'Location', 'northwest');
91 grid on;
```

Part (a) - Firefly and Cricket Analysis:

Similarities: Both show increasing activity with temperature

Differences: Cricket rates much higher (80-200 vs 9-16), different temp ranges

Part (b) - Honeybee Velocities:

Velocity at 40°C: 4.30 cm/s

Velocity at -5°C: -3.80 cm/s

Part (c) - Common patterns:

All insects show positive temperature-activity correlation

Similar exponential-type growth with temperature

Part (d) - Experimental improvements:

Need wider temperature ranges and more data points

Suggest testing ants: easy to observe, measure running speed

Temperature range: 5-45°C in 5°C steps

Part (e) - Beetle analysis:

Predicted speed at 41.5°C: 21.0 cm/s

Reference value: 19.2 cm/s

>> ✦ Press **Ctrl** + **Shift** + **P** to generate code with Copilot